

Machine Learning - Mid-Semester 1 Exam Guide

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Machine Learning Midsem 1 - Six-Day Preparation Hub

Course: AMLCS ZG565 Machine Learning

User-provided regular-exam schedule (not independently verified): 26 September 2026, 01:00 PM–03:00 PM IST ([schedule](#))

Scope: Contact Sessions 1–8; closed book. The handout specifies two hours, while the practice mock intentionally uses the repository's common 90-minute evaluation protocol.

Accessibility: The generated PDFs are currently untagged and may not expose reliable semantic structure to assistive technology. Use these linked Markdown files as the accessible source alternative.

Six-day route

Day	Learn	Practise	Check
1	Session 1 , Session 2 , and cards 1, 2	C01–C07 , N01–N07	Explain T–E–P, leakage-safe splitting, and metrics without notes.
2	Session 3 and card 3	C08–C11 and N08–N11	Recalculate normal equations and one simultaneous gradient step.
3	Session 4 , Session 5 , and cards 4, 5	C12–C17 and N12–N17	Separate scores, distances, probabilities, decisions, and risks.
4	Session 6 and card 6	C18–C20 and N18–N21	Calculate entropy, Gini, and weighted gain from counts.
5	Session 7 , Session 8 , and cards 7, 8	C21–C24 , N22–N24 , and BITS-style practice	Contrast kNN, LWR, RBF, selection, and final evaluation.
6	Sit Mock 01 in 90 minutes	Mark only after submission with the answer key and rubric	Finish with the last-30-minutes route .

Non-negotiable answer pattern

For every numerical or applied answer: state the formula, define symbols and shapes, state conditions, substitute values, show arithmetic, and interpret the result. For a qualitative scenario, state assumptions and justify the decision.

Machine Learning Midsem 1 - Standalone Exam Guide

The handout prescribes Sessions 1–8. “Local evidence” below reports only whether the available recordings/transcripts substantively cover a session; it does not determine technical truth. Sessions 6–8 lack substantive local lecture evidence, so their coverage here comes from the handout-prescribed scope and the cited technical sources without claiming classroom delivery.

Session	Handout-prescribed scope	Local teaching evidence
1	Introduction, objectives, applications, system design, issues	Substantive: 08-08 and 05-09
2	Mathematical preliminaries, workflow, preprocessing, sampling, training/testing, metrics	Substantive: 16-08 and 05-09
3	Regression, direct/iterative methods, basis functions, bias–variance	Substantive: 22-08 and 29-08
4	Discriminants, decision theory, probabilistic discriminative classifiers, logistic introduction	Substantive: 30-08
5	Log loss, gradient descent, multiclass classification	Partial: binary logistic only on 30-08; multiclass not evidenced
6	Decision trees, information theory, overfitting, MDL, continuous/missing values	Not substantively evidenced locally
7	kNN, locally weighted regression, radial basis functions	Introductory instance-based mention only; prescribed detail not evidenced
8	Review of Sessions 1–7	Not substantively evidenced locally

Prerequisite refreshers

- A scalar is one number. A vector is an ordered column of numbers. A matrix X with n rows and d columns has shape n by d .
- X^T swaps rows and columns. If X is n by d and w is d by 1 , then Xw is n by 1 .
- A derivative measures local slope. A gradient lists one partial derivative per fitted parameter and has the same shape as that parameter vector.
- Probability lies from 0 to 1. Conditional probability $P(A|B)$ means the probability of A after learning B .

- $\ln$ is natural logarithm; $\log_2$ measures information in bits. The exponential $\exp(z)$ is always positive.

Session 1 - Introduction

What counts as learning

A program learns from experience E with respect to task T and performance measure P when its measured performance at T improves with E . This forces a vague project into testable parts. For a house-price system, T is predicting price, E is past houses with sale prices, and P might be mean absolute error on future-like houses.

Supervised learning uses target labels. Regression targets numbers; classification targets classes. Unsupervised learning seeks structure without target labels. Semi-supervised learning combines limited labels with unlabeled examples. Reinforcement learning learns a policy—an action rule—from interaction and rewards.

Designing the system

- 1 Define the decision and success measure.
- 2 Identify inputs available at prediction time and a representative target.
- 3 Choose a hypothesis representation and learning procedure.
- 4 Protect an evaluation protocol before fitting.
- 5 Examine failures, costs, drift, and deployment constraints.

Common issues are biased sampling, weak labels, leakage, mismatch between training and deployment, overfitting, and a metric that rewards the wrong behavior. Correlation alone does not establish that changing a feature causes an outcome.

Session 2 - Mathematical Preliminaries and Workflow

Data and safe preprocessing

Numeric features may be interval/ratio measurements; categorical features may be nominal or ordered. Encoding changes representation, not meaning. Split first. Fit imputation, scaling, selection, and learned encoding only on training data; apply frozen transformations to validation, test, and deployment inputs.

For a training mean μ and standard deviation $s > 0$, standardization is $z = (x - \mu)/s$. It does not require Gaussian data. For training extrema with $\max > \min$, min-max scaling is $(x - \min)/(\max - \min)$. A constant training feature creates a zero denominator; remove it or use a documented finite convention.

Tukey's inner fences are $Q1 - 1.5 \text{ IQR}$ and $Q3 + 1.5 \text{ IQR}$, where $\text{IQR} = Q3 - Q1$. Crossing a fence flags an observation; it does not prove an error. The statement that about 99.73% lies within three standard deviations applies to a normal distribution.

Splits and metrics

Training fits parameters. Validation selects model family, hyperparameters, preprocessing choices, and threshold. Test estimates performance after all choices are frozen. Time-dependent and grouped data often require time/group-aware splits rather than random rows.

For binary counts:

accuracy	precision	recall	F1
$\frac{TP + TN}{TP + TN + FP + FN}$	$\frac{TP}{TP + FP}$	$\frac{TP}{TP + FN}$	$\frac{2PR}{P + R}$

Each denominator must be nonzero; P means precision and R means recall.

- true positive (TP) is a correctly predicted positive; false positive (FP) is a negative example incorrectly predicted positive; true negative (TN) is a correctly predicted negative; false negative (FN) is a positive example incorrectly predicted negative;
- accuracy $(TP+TN)/(TP+TN+FP+FN)$;
- precision $TP/(TP+FP)$ asks how many predicted positives are correct;
- recall $TP/(TP+FN)$ asks how many actual positives were found;
- F1 is $2(\text{precision})(\text{recall})/(\text{precision}+\text{recall})$ when its denominator is nonzero.

The right metric follows the cost of errors. Accuracy can be misleading for an imbalanced rare class.

Session 3 - Linear Models for Regression

Model, loss, and direct solution

Let X be n by d , including a constant column if an intercept is fitted. Let w be d by 1 and target y be n by 1 . Prediction is $\hat{y} = Xw$. Least squares minimizes $J(w) = (1/n) \|Xw - y\|^2$.

Its gradient is $(2/n)X^T(Xw - y)$. Setting it to zero gives the normal equations $X^T Xw = X^T y$. The compact inverse expression requires X to have full column rank and the Gram matrix to be acceptably conditioned. Stable least-squares factorizations or the pseudoinverse are safer when columns are dependent or nearly dependent.

Gradient descent and basis functions

Batch gradient descent uses all n rows per step; stochastic uses one; mini-batch uses a subset. For batch MSE:

$$w(\text{new}) = w(\text{old}) - \alpha \frac{2 X^T (Xw(\text{old}) - y)}{n}$$

Batch MSE gradient step

$X: n \text{ by } d \cdot w: d \text{ by } 1 \cdot y: n \text{ by } 1 \cdot n > 0 \cdot \alpha > 0$

All components must be computed from the same old w . $\alpha > 0$ is the learning rate. Scaling affects the geometry and hence practical convergence.

A polynomial basis $\phi(x) = (1, x, x^2, \dots)$ makes predictions nonlinear in x while remaining linear in weights. Greater flexibility can reduce training error but raise variance. Select degree or regularization on validation data. Ridge adds an L2 squared penalty. Lasso adds L1 absolute values; it is nonsmooth at zero but optimizable.

$R^2 = 1 - \text{SSE}/\text{SST}$ compares residual error with variation around the observed mean. It needs $\text{SST} > 0$; a constant target makes the elementary ratio undefined.

Session 4 - Linear Models for Classification

A linear discriminant $g_k(x)$ is a class score; choose the largest score. Binary $g(x) = w^T x + b$ creates boundary $g(x) = 0$. Its sign gives the side. Signed geometric distance is $g(x)/\|w\|_2$ only when w is nonzero; raw magnitude changes if both w and b are rescaled.

A generative classifier models $P(x|C_k)$ and priors $P(C_k)$, then uses Bayes' rule. A discriminative classifier directly models $P(C_k|x)$ or a decision boundary. Neither label alone guarantees superior performance.

Decision theory separates belief from action. If action a_i incurs loss $\lambda(a_i|C_j)$ when the true class is C_j , conditional risk is:

$$R(a_i|x) = \sum_j \lambda(a_i|C_j) P(C_j|x)$$

Choose the action with minimum risk. Maximum-posterior classification is a special case under equal misclassification costs.

Session 5 - Logistic Regression

Binary logistic regression uses linear score $z = w^T x + b$ and probability $p = 1/(1 + \exp(-z))$. Its log-odds satisfy $\ln(p/(1-p)) = z$. For label y in $\{0, 1\}$, one-example cross-entropy is:

$$L = -[y \ln(p) + (1-y) \ln(1-p)]$$

For n rows, the average gradient with respect to w is $(1/n)X^T(p-y)$. Use stable log-loss routines because $\ln(0)$ is undefined. Threshold selection is a validation decision separate from fitting. Probabilities are not guaranteed calibrated.

For K mutually exclusive classes, softmax converts logits $z_1, \dots, z_K$ to $p_k = \exp(z_k) / \sum_j \exp(z_j)$. The probabilities are positive and sum to 1. With one-hot target y , cross-entropy is $-\sum_k y_k \ln(p_k)$. Subtract the largest logit before exponentiation for numerical stability.

softmax	cross-entropy
$p_k = \frac{\exp(z_k)}{\sum_j \exp(z_j)}$	$L = -\sum_k y_k \ln(p_k)$
	one-hot target: $L = -\ln(p_t)$

Linear binary or multinomial logistic cross-entropy with a convex $L1/L2$ penalty is convex. This statement does not extend to arbitrary neural networks containing sigmoid or softmax units.

Session 6 - Decision Trees

A tree recursively partitions feature space. A classification leaf stores a class decision or class proportions. For proportions p_k , entropy is $H = -\sum p_k \log_2(p_k)$ with $0 \log 0 = 0$, while Gini impurity is $G = 1 - \sum p_k^2$. Both are zero for a pure node. Balanced binary data give entropy 1 bit and Gini 0.5.

For children containing n_c of n parent examples, impurity reduction is:

node impurity

$$H = -\sum_k p_k \log_2(p_k)$$

$$G = 1 - \sum_k p_k^2$$

split gain

$$\text{gain} = I(\text{parent}) - \sum_c w_c I(c)$$

$$w_c = \frac{n_c}{n}$$

Use $0 \log 0 = 0$; child weights sum to one.

$$\text{gain} = I(\text{parent}) - \sum_c (n_c/n) I(\text{child}_c)$$

For a continuous feature, candidate questions $x_j \leq t$ versus $x_j > t$ are evaluated at thresholds; prior discretization is not universally required. Missing values require an explicit policy. Pre-pruning limits depth or leaf size. Post-pruning removes branches using a fit-complexity trade-off selected on validation data.

candidate continuous split



Search candidate thresholds; prior binning is not required.

Minimum Description Length chooses the representation minimizing the combined coded length of the model and residual exceptions under a specified coding scheme. It is not simply “fewest nodes.”

Checked split: parent (4+,4-) has entropy 1. Children (3+,1-) and (1+,3-) each have entropy 0.8113; equal weighting gives gain $1 - 0.8113 = 0.1887$ bit.

Session 7 - Instance-Based Learning

kNN stores examples, computes a distance from query q , selects k , and votes or averages. Euclidean distance is $\sqrt{\sum_j (x_j - q_j)^2}$. Scaling matters because units control squared contributions. Choose k , metric, scaling, weighting, and tie rules using validation data.

Euclidean distance

$$d(x, q) = \sqrt{\sum_j (x_j - q_j)^2}$$

inspect units before comparing coordinates

weighted class vote

$$\text{score}(C) = \frac{1}{\sum d_i}$$

sum only neighbours with label C
requires $d_i > 0$ or an explicit exact-match rule

Locally weighted regression fits a new weighted regression near every query. A common weight is $a_i = \exp(-||x_i - q||^2 / (2 \tau^2))$, $\tau > 0$. Small bandwidth is flexible and variable; large bandwidth is smoother and more biased.

LWR weight near query q

$$a_i = \exp\left(-\frac{\|x_i - q\|_2^2}{2\tau^2}\right)$$

$\tau > 0$; fit a weighted regression separately for each query.

An RBF feature around centre c_j is $\phi_j(x) = \exp(-\|x - c_j\|_2^2 / (2s_j^2))$, $s_j > 0$. A linear combination of RBF features is a global fitted model, unlike kNN voting and query-specific LWR.

RBF feature around centre c_j

$$\phi_j(x) = \exp\left(-\frac{\|x - c_j\|_2^2}{2s_j^2}\right)$$

$s_j > 0$; global coefficients combine the resulting features.

Checked vote: labels A,A,B at distances 2,4,1 give uniform class A; inverse-distance weights give $A=0.75$, $B=1$, hence class B. Exact zero distance needs a declared rule rather than division by zero.

Session 8 - Integrated Review

For any scenario, answer this chain:

- 1 Is the target numeric, categorical, or absent?
- 2 What information is available at prediction time?
- 3 Which split respects time, groups, and deployment?
- 4 Which preprocessing is fitted only on training data?
- 5 Which baseline and candidate model match the output and representation?
- 6 Which loss fits parameters, which validation metric selects choices, and which threshold implements costs?
- 7 After freezing every choice, what one-time test result supports the claim?

Mixed checked practice

- 1 Confusion counts $TP=18, FP=6, FN=2, TN=24$ give accuracy $42/50=0.84$, precision $18/24=0.75$, and recall $18/20=0.90$.
- 2 For logits $(0, \ln 2, \ln 3)$, softmax is $(1/6, 1/3, 1/2)$; true third-class loss is $\ln 2$.
- 3 A pure tree child has entropy and Gini zero, but this alone does not prove the full tree generalizes.
- 4 A fraud model tuned repeatedly on the newest period has consumed its test set; obtain a fresh untouched period for an honest final estimate.

Formula recall and common failure checklist

- Define every symbol, shape, class convention, and positive denominator.

- Never call an IQR flag proof, raw classifier score a distance/confidence, or sigmoid output automatically calibrated.
- State full-rank/conditioning limits for the inverse least-squares form and $SST > 0$ for R^2 .
- Fit preprocessing and select models/thresholds without touching the final test set.
- For trees, weight child impurity by child size. For distance methods, inspect units and zero-distance cases.

External Sources

Conceptual Drills - Answer-Free

For each response, define new terms, state conditions, and justify the conclusion. These 24 prompts cover only Sessions 1–8.

- 1 C01 (S1): Define learning through task, experience, and performance; construct all three for predictive maintenance.
- 2 C02 (S1): Distinguish supervised, unsupervised, semi-supervised, and reinforcement learning by the feedback available.
- 3 C03 (S1): Explain why a representative experience and aligned performance measure are part of system design.
- 4 C04 (S2): State the separate roles of training, validation, and test sets and explain test-set “wear-out.”
- 5 C05 (S2): Explain why preprocessing fitted before splitting is leakage, using scaling as the example.
- 6 C06 (S2): Contrast a loss, metric, and decision threshold.
- 7 C07 (S2): State precisely what an IQR fence and a three-sigma interval do—and do not—establish.
- 8 C08 (S3): Define prediction, residual, least-squares objective, coefficient, and basis function.
- 9 C09 (S3): State the rank and conditioning assumptions behind the inverse normal-equation expression.
- 10 C10 (S3): Contrast batch, stochastic, and mini-batch gradient descent.
- 11 C11 (S3): Diagnose high bias and high variance using training and validation errors.
- 12 C12 (S4): Contrast discriminant score, signed geometric distance, posterior probability, and hard decision.
- 13 C13 (S4): Contrast generative and discriminative classification without claiming either always wins.
- 14 C14 (S4): Derive the minimum-conditional-risk decision rule from a loss table and class posteriors.
- 15 C15 (S5): Explain sigmoid, odds, log-odds, and the interpretation of a logistic coefficient.
- 16 C16 (S5): Contrast binary sigmoid, one-vs-rest, and multinomial softmax.
- 17 C17 (S5): State the exact scope and limitations of logistic-regression convexity.
- 18 C18 (S6): Define node impurity, entropy, Gini impurity, and information gain.
- 19 C19 (S6): Explain tree overfitting, pre-pruning, post-pruning, and validation's role.

- 20 C20 (S6): Explain why continuous attributes need not be pre-binned and why missing values need an explicit policy.
- 21 C21 (S7): Contrast kNN classification, kNN regression, LWR, RBF models, and k-means.
- 22 C22 (S7): Explain how k , bandwidth, scaling, and dimension affect bias and variance.
- 23 C23 (S8): Design an end-to-end leakage-safe workflow for time-ordered fraud detection.
- 24 C24 (S8): Explain why best training loss, best validation metric, and final test estimate answer different questions.

Numerical Drills - Answer-Free

Show formulas, substitutions, intermediate arithmetic, conditions, shapes, and interpretation. Use natural logarithms unless a prompt says $\log_2$.

- 1 N01 (S1): A defect detector improves recall from 0.70 to 0.84 after learning from 2,000 labeled images. Identify T , E , P , and the absolute and relative recall improvement.
- 2 N02 (S1): A classifier makes 92 correct decisions in 100 trials, but the cost is 1 for each false positive and 20 for each false negative. Given $FP=6$, calculate FN and total error cost; explain why accuracy alone is incomplete.
- 3 N03 (S1): A model's error is 18 before experience and 12 after. For a lower-is-better measure, calculate absolute and percentage error reduction.
- 4 N04 (S2): Given training values 2,4,6,8, compute the population training mean and standard deviation, then standardize a test value 10 using only those values.
- 5 N05 (S2): Training min/max are 5,25. Scale validation value 15; then state what the elementary formula does if training min=max.
- 6 N06 (S2): For $TP=30, FP=10, FN=20, TN=40$, compute accuracy, precision, recall, and $F1$.
- 7 N07 (S2): For sorted values 1,2,3,4,5,6,7,20 with quartiles supplied as $Q1=2.5$, $Q3=6.5$, compute IQR and inner fences and classify 20 as flagged/not flagged.
- 8 N08 (S3): For X with rows (1,0),(1,1),(1,2) and $y=(1,3,5)$, solve the two normal equations for intercept and slope.
- 9 N09 (S3): With X rows (1,1),(1,2), $y=(2,3)$, $w=(0,0)$, and MSE gradient $(2/n)X^T(Xw-y)$, compute one simultaneous step at $\alpha=0.1$.
- 10 N10 (S3): Observed $y=(1,2,3)$ and predicted $\hat{y}=(1,2,4)$. Compute SSE, MSE, SST, and R^2 ; state the SST condition.
- 11 N11 (S3): For $\phi(x)=(1,x,x^2)$ and $w=(1,2,-1)$, compute the prediction at $x=3$ and identify all shapes.
- 12 N12 (S4): With $w=(3,4)$, $b=-10$, and $x=(2,1)$, compute score, predicted side under $g \geq 0$, and signed geometric distance.
- 13 N13 (S4): Posterior disease probability is 0.15; false-treatment cost is 2, missed-disease cost is 20, correct costs zero. Compute both conditional risks and choose the action.
- 14 N14 (S4): Two class scores are $g_1=2x_1-x_2$ and $g_2=-x_1+2x_2$. At $x=(3,1)$, compute both and predict by argmax .

- 15 N15 (S5): For logistic score $z=\ln 3$, compute odds and probability; at threshold 0.8, state the hard class.
- 16 N16 (S5): For $y=1, p=0.8$, compute binary log loss; repeat for $y=0, p=0.8$.
- 17 N17 (S5): For logits (0, $\ln 2$, $\ln 3$) and true class 3, compute softmax probabilities and cross-entropy.
- 18 N18 (S6): Compute entropy and Gini for binary counts (3, 1).
- 19 N19 (S6): A parent has (4+, 4-); children are (3+, 1-) and (1+, 3-). Compute entropy information gain.
- 20 N20 (S6): A parent Gini is 0.48; children of sizes 3 and 7 have Gini 0.20 and 0.40. Compute weighted child impurity and gain.
- 21 N21 (S6): Sorted feature values and labels are (1,A),(2,A),(4,B),(7,B). For thresholds 3 and 5.5, compute weighted Gini and identify the better split.
- 22 N22 (S7): Query $q=(2,2)$ has neighbours A:(2,4), A:(5,2), B:(2,1). Compute Euclidean distances, uniform $k=3$ vote, and inverse-distance vote.
- 23 N23 (S7): At squared distances 0, 2, 8 and bandwidth $\tau=2$, compute Gaussian LWR weights $\exp(-d^2/(2\tau^2))$ to three decimals.
- 24 N24 (S8): Two validation models give (precision, recall)=(0.8, 0.5) and (0.6, 0.9). Compute F1 for both and select under F1; then explain why the final test set still remains untouched.

BITS-Style Practice - Answer-Free

No authentic matching BITS examination paper was available from an official source, so none of these items claims historical status.

H01 - Authored BITS-style

Classification: Authored BITS-style | Institution: Birla Institute of Technology & Science, Pilani - WILP | Code: target exact AMLCS ZG565; historical code not applicable | Term/year: authored for Midsem 1 preparation, 2026; historical term/year not applicable | Hosting: official bulletin, but no historical question claim | Source: [BITS Bulletin 2025-26](#)

A medical-screening model gives $TP=72$, $FP=18$, $FN=8$, $TN=102$. Compute accuracy, precision, recall, and F1. If missing one illness case costs five times a false alarm, state which metric deserves special attention and why.

H02 - Authored BITS-style

Classification: Authored BITS-style | Institution: Birla Institute of Technology & Science, Pilani - WILP | Code: target exact AMLCS ZG565; historical code not applicable | Term/year: authored for Midsem 1 preparation, 2026; historical term/year not applicable | Hosting: official bulletin, but no historical question claim | Source: [BITS Bulletin 2025-26](#)

A binary tree node has counts (6+, 4-). Candidate split A creates (4+, 0-) and (2+, 4-); split B creates (3+, 1-) and (3+, 3-). Compute entropy information gain for both and select one, stating the $0 \log 0$ convention.

H03 - Authored BITS-style

Classification: Authored BITS-style | Institution: Birla Institute of Technology & Science, Pilani - WILP | Code: target exact AMLCS ZG565; historical code not applicable | Term/year: authored for Midsem 1 preparation, 2026; historical term/year not applicable | Hosting: official bulletin, but no historical question claim | Source: [BITS Bulletin 2025-26](#)

Design a leakage-safe comparison of logistic regression, a depth-limited tree, and kNN for time-stamped loan-default data. State split order, preprocessing ownership, validation choices, metric/cost reasoning, and the single final test action.